

Adding and Removing a Belief

A linear-algebra picture of Bayesian model reduction and expansion,
and why you can rotate one object to see both

Abstract

We build a single mathematical object — the *information matrix* of a belief — and then rotate it. Seen one way it tells you how to *add* a theory to a model (border the matrix); rotated, it tells you how to *remove* one (Schur-complement the matrix); and the price of either move, in units of model evidence, is read off the same function, the moment generating function of the posterior. Along the way we say plainly why the inversion that Bayes seems to demand is hard, what trick makes it cheap, and how the whole thing moves in time as three rival theories A, B, C chase a changing world. We close by mapping every pen stroke of the “Bayesian Model Expansion & Reduction” sketch onto the algebra.

1 The one object

A Gaussian belief about a vector of parameters $\varphi \in \mathbb{R}^d$ can be written in two ways. The familiar way is by its mean μ and covariance Σ . The useful way — the way that makes everything in this note fall out — is by its *information*: the precision matrix $\Pi = \Sigma^{-1}$ and a potential vector $h = \Pi\mu$,

$$p(\varphi) \propto \exp\left(-\frac{1}{2}\varphi^\top \Pi \varphi + h^\top \varphi\right). \quad (1)$$

Read Π as a graph. The diagonal entry Π_{ii} is how strongly you believe in parameter i on its own; the off-diagonal Π_{ij} is how tightly your belief about i is yoked to your belief about j . Zero off-diagonal means conditional independence — no edge. A dense block is a tightly-coupled core; an isolated near-diagonal row is a belt node that floats almost free (Figure 1). *The precision matrix is the paradigm net, written as numbers.*

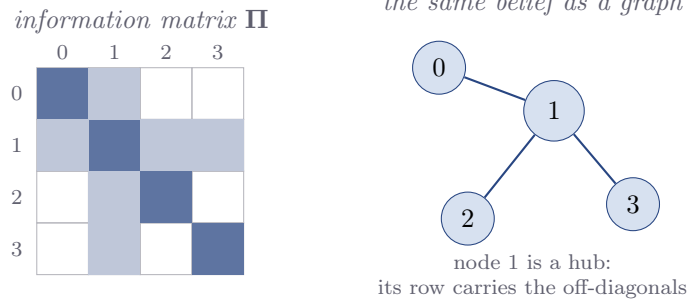


Figure 1: The precision (information) matrix Π is the belief network. Dark diagonal = self-precision; light off-diagonal = inferential coupling (an edge). A zero off-diagonal is a missing edge (conditional independence). Node 1 here is a hub — a core node — because its row is full.

Why this form and not mean–covariance? Because in this form *evidence is additive and combination is addition*. If two independent sources of information about φ have parameters (Π_1, h_1) and (Π_2, h_2) , the combined belief is

$$\Pi = \Pi_1 + \Pi_2, \quad h = h_1 + h_2. \quad (2)$$

Multiplying probability densities became *adding* matrices. Hold onto that — it is the answer to “when do I add something to the model.” You add to the model by adding to $\mathbf{\Pi}$.

Slogan. A belief is an information matrix. To learn is to add to it. To restructure is to border it or to Schur-complement it. To price the restructuring is to evaluate a moment generating function.

2 Three operations, not one: learn, expand, reduce

The thing that tangles people — you said this exactly — is not knowing *when* you add to a model and when you remove. There are really three distinct operations, and confusing them is the whole difficulty. Let me separate them cleanly. Partition the parameters into a block we keep, φ_a , and a block we are deciding about, φ_b , so that

$$\mathbf{\Pi} = \begin{pmatrix} \mathbf{\Pi}_{aa} & \mathbf{\Pi}_{ab} \\ \mathbf{\Pi}_{ba} & \mathbf{\Pi}_{bb} \end{pmatrix}, \quad \mathbf{h} = \begin{pmatrix} \mathbf{h}_a \\ \mathbf{h}_b \end{pmatrix}. \quad (3)$$

(1) Learn — add data. Each observation deposits a little information matrix J_τ (its Fisher information) and a potential $\mathbf{j}_\tau$. By Eq. (2) the posterior after t observations is just a running sum,

$$\mathbf{\Pi}(t) = \mathbf{\Pi}_0 + \sum_{\tau \leq t} J_\tau, \quad \mathbf{h}(t) = \mathbf{h}_0 + \sum_{\tau \leq t} \mathbf{j}_\tau. \quad (4)$$

Learning does not change the *size* of the matrix, only its entries. This is the dynamics; we return to it in §7.

(2) Expand — add a parameter (your “new” node). Now we add a genuinely new degree of freedom φ_b — a new theoretical commitment, a new node connected to the existing network. The matrix *grows*: we *border* it with a new row and column whose entries are the couplings $\mathbf{\Pi}_{ab}$ of the newcomer to the incumbents (Figure 2). The newcomer’s conditionals $p(\text{node } k \mid \text{new})$ in your sketch are exactly these off-diagonal blocks.

(3) Reduce — remove a parameter. Here is the subtlety that trips everyone: there are *two* ways to remove φ_b , and they are not the same.

- **Marginalize** (integrate it out). You keep believing the newcomer might matter, but you no longer track it explicitly; you push its information into correlations among the survivors. This is the *Schur complement* (§3).
- **Prune** (assert it is absent). You sharpen its prior to a spike — declare the coupling zero — and ask the data whether that assertion costs you evidence. This is *Bayesian model reduction* proper (§4), and the cost is the moment generating function (§5).

Marginalizing redistributes; pruning amputates and reads the bill. Most of the confusion about BMR is really confusion between these two “removes.”

3 Removing by marginalizing: the Schur complement is the carry-over

Suppose we decide to stop tracking φ_b explicitly and integrate it out. For a Gaussian this is exact, and in information form the answer is the *Schur complement*: the surviving block keeps

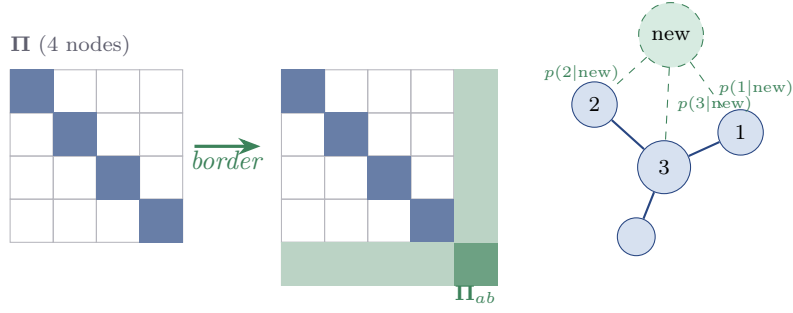


Figure 2: **Expansion by bordering.** Adding a new commitment grows Π by one row and column (green). The new diagonal is how strongly the newcomer is believed on its own; the new off-diagonal block Π_{ab} holds its couplings to the incumbents — exactly the conditionals $p(k | \text{new})$ in the sketch. The dimension increases; no incumbent entry is touched.

precision

$$\Pi_a^{\text{marg}} = \underbrace{\Pi_{aa}}_{\text{direct}} - \underbrace{\Pi_{ab} \Pi_{bb}^{-1} \Pi_{ba}}_{\text{carry-over fill-in}}, \quad h_a^{\text{marg}} = h_a - \Pi_{ab} \Pi_{bb}^{-1} h_b. \quad (5)$$

Look at what the subtracted term does. Before, nodes that were coupled only *through* φ_b had no direct edge. After we remove φ_b , the term $\Pi_{ab} \Pi_{bb}^{-1} \Pi_{ba}$ writes new off-diagonal entries among exactly those nodes (Figure 3). The removed node leaves its ghost behind as fresh couplings among its former neighbours. *This is the carry-over property, in one line of linear algebra.* You cannot delete a hub cleanly; its influence re-precipitates as correlations among everything it touched, and the more it touched — the larger Π_{ab} — the denser the ghost.

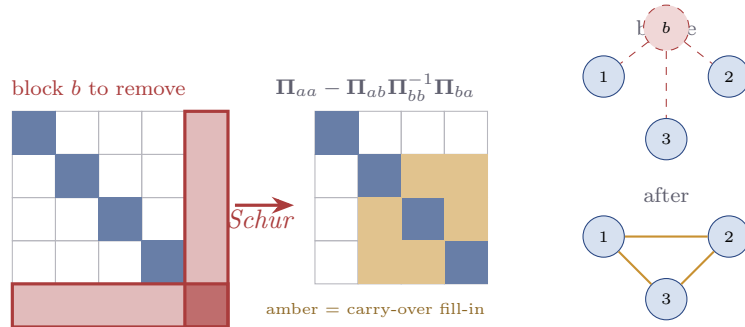


Figure 3: **Reduction by marginalizing is the Schur complement.** Integrating out block b subtracts $\Pi_{ab} \Pi_{bb}^{-1} \Pi_{ba}$ from the survivors, which *writes new edges* (amber) among b 's former neighbours. The deleted node's influence does not vanish; it re-precipitates as carry-over coupling. A hub leaves a dense ghost.

Two facts are worth saying out loud because they are the source of half the practical pain.

First, a duality. In *covariance* form marginalizing is trivial (delete rows) and conditioning is hard; in *information* form it is exactly reversed — conditioning is trivial (delete rows of Π) and marginalizing costs you the Schur complement. The two representations are the same object rotated; whichever you pick, one of the two removals is cheap and the other is the inverse you were hoping to avoid.

Second, the Schur complement contains a matrix inverse, Π_{bb}^{-1} . If b is a single scalar this is a division; if b is a whole sub-network it is a genuine inversion. That is the first place the “inverse Bayes is hard” problem bites, and §6 is about why it usually does not bite as hard as it looks.

4 Removing by pruning: Bayesian model reduction

Marginalizing answers “what do I believe about the survivors if I stop tracking b .” It does *not* answer “should I have b at all.” For that we need the model *evidence*, and the beautiful fact — the whole of BMR — is that we can get the evidence of any *pruned* model from the full posterior we already computed, with no re-fitting.

A pruned model has the *same likelihood* but a different (sharper) prior $\tilde{p}(\boldsymbol{\varphi})$ — typically one that pins a coupling to zero. Start from the two Bayes rules, solve the full one for the shared likelihood, substitute into the reduced one, and integrate. In two lines (this is Friston & Penny’s post-hoc identity, the matrix generalization of the Savage–Dickey density ratio):

$$\boxed{\frac{\tilde{\mathcal{Z}}}{\mathcal{Z}} = \mathbb{E}_{p(\boldsymbol{\varphi}|y)} \left[\frac{\tilde{p}(\boldsymbol{\varphi})}{p(\boldsymbol{\varphi})} \right]} \quad \text{and} \quad \tilde{p}(\boldsymbol{\varphi} | y) \propto p(\boldsymbol{\varphi} | y) \frac{\tilde{p}(\boldsymbol{\varphi})}{p(\boldsymbol{\varphi})}. \quad (6)$$

The Bayes factor for any structural edit is the posterior expectation of the prior ratio. Everything on the right is already in hand. No likelihood is re-evaluated; the posterior has already drunk the data, and a prior change merely re-weights what it drank.

The sign of $\log(\tilde{\mathcal{Z}}/\mathcal{Z})$ is the verdict. Sharpen the prior on a coupling toward a spike at zero: if the data are content with the coupling absent, the ratio exceeds one ($\log > 0$) and you prune; if the data hold the coupling away from zero, the ratio falls below one and you keep it. This is automatic Occam: the complexity term of the free energy, $D_{\text{KL}}(p(\boldsymbol{\varphi} | y) \parallel p(\boldsymbol{\varphi}))$, is exactly the price a parameter pays for the prior volume it occupies, and pruning refunds that price whenever the data did not need the volume.

5 The ledger is a moment generating function

Now to the question you really asked: in density language, *what is the act of adding or removing?* It is *exponential tilting*, and the bill is the *moment generating function*. Here is the cleanest case.

Multiply a belief $p(\boldsymbol{\varphi})$ by an exponential factor $e^{\mathbf{t}^\top \boldsymbol{\varphi}}$ — this is what “adding a linear piece of evidence” or “adding a coupling to a neighbour” does to the density. The new normalizer is, by definition,

$$\mathcal{Z}(\mathbf{t}) = \int p(\boldsymbol{\varphi}) e^{\mathbf{t}^\top \boldsymbol{\varphi}} d\boldsymbol{\varphi} = \mathbb{E}_p[e^{\mathbf{t}^\top \boldsymbol{\varphi}}] = M_p(\mathbf{t}), \quad (7)$$

the *moment generating function* of p evaluated at $\mathbf{t}$. To *add* the factor is to multiply the evidence by $M_p(\mathbf{t})$; to *remove* it is to divide by $M_p(\mathbf{t})$ — tilt, and untilt. The logarithm $K_p(\mathbf{t}) = \log M_p(\mathbf{t})$ is the cumulant generating function, and for a Gaussian it is the tidy quadratic

$$K_p(\mathbf{t}) = \boldsymbol{\mu}^\top \mathbf{t} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}, \quad (8)$$

whose first derivative is the mean and whose second is the covariance — the moments, generated. Compare Eq. (6): the BMR Bayes factor is $\mathbb{E}_{p(\boldsymbol{\varphi}|y)}[\tilde{p}/p]$, and when the prior ratio is an exponential-family tilt $\tilde{p}/p \propto e^{\mathbf{t}^\top \boldsymbol{\varphi}}$ (a mean-shift reduction), this expectation *is* the posterior’s MGF,

$$\frac{\tilde{\mathcal{Z}}}{\mathcal{Z}} = M_{p(\cdot|y)}(\mathbf{t}). \quad (9)$$

Slogan. To add a belief is to tilt the distribution by an exponential; to remove it is to tilt back. The evidence you gain or pay is the moment generating function of what you already believe, read at the tilt. The moment generating function is the ledger of structural change.

A pure precision change (pin a coupling, not just shift a mean) is the generalized version: the prior ratio carries a quadratic $e^{-\frac{1}{2} \boldsymbol{\varphi}^\top \Delta \boldsymbol{\Pi} \boldsymbol{\varphi}}$, and $\mathbb{E}_{p(\cdot|y)}[\cdot]$ becomes the Gaussian integral of a

posterior against a quadratic tilt — still an MGF-type object, now of the squared parameter. The determinant and Mahalanobis terms in the closed-form ΔF are precisely the log of that integral. So: mean-shift edits are read off the MGF; precision edits are read off its Gaussian generalization; both are the partition function Eq. (7) doing its job.

6 Why the inverse is hard, and the trick that saves you

Bayes asks for a normalizer, $\mathcal{Z} = \int p(y | \varphi) p(\varphi) d\varphi$, and for a Gaussian that means $\mathcal{Z} \propto |\mathbf{\Pi}|^{-1/2} \exp(\frac{1}{2} \mathbf{h}^\top \mathbf{\Pi}^{-1} \mathbf{h})$ — you need $\mathbf{\Pi}^{-1}$, a $d \times d$ inverse, cost $O(d^3)$. Worse, naively scoring every candidate sub-model means re-inverting a different matrix each time: 2^K structural edits, each an $O(d^3)$ inversion. That is the wall people mean when they say inverse Bayesian modelling is hard.

The escape is that structural edits are *low-rank*. Pruning or adding one edge changes $\mathbf{\Pi}$ by a rank-one (or low-rank) bump $\mathbf{\Pi} \mapsto \mathbf{\Pi} + \mathbf{u}\mathbf{v}^\top$, and the Sherman–Morrison–Woodbury identity turns a low-rank change of a matrix into a low-rank change of its *already-computed* inverse,

$$(\mathbf{\Pi} + \mathbf{UCV})^{-1} = \mathbf{\Pi}^{-1} - \mathbf{\Pi}^{-1} \mathbf{U} (\mathbf{C}^{-1} + \mathbf{V} \mathbf{\Pi}^{-1} \mathbf{U})^{-1} \mathbf{V} \mathbf{\Pi}^{-1}. \quad (10)$$

You pay the $O(d^3)$ inversion *once*, for the full model; thereafter each candidate edit is an $O(d^2)$ update, or cheaper for a single edge. Schur complement (Eq. 5) and Woodbury (Eq. 10) are two faces of the same block-elimination identity — the first removes a block, the second patches the inverse when a block changes. *BMR is exactly the discipline of inverting once and updating forever.* That is why it makes searching over paradigm topologies feasible at all: model search collapses from “solve 2^K inverse problems” to “solve one, then evaluate 2^K algebraic expressions.”

7 The dynamics: three theories chasing a moving world

Now set it in motion. You have three candidate theories A, B, C . Each is a generative model; each carries its own running information $(\mathbf{\Pi}^M(t), \mathbf{h}^M(t))$ accumulating by Eq. (4); each has a running log-evidence $\log \mathcal{Z}^M(t)$. Two facts give the whole dynamical picture.

Drift. The expected one-step gain in a model’s log-evidence, under data from the true process P , is

$$\mathbb{E}_P[\log p_M(o)] = -H(P) - D_{\text{KL}}(P \| p_M), \quad (11)$$

so the *rate* at which $\log \mathcal{Z}^M$ climbs is set, up to a common constant, by $-D_{\text{KL}}(P \| p_M)$: the model closest to the truth has the steepest drift and pulls ahead. The running log-evidence of each theory is a random walk with drift, and the drifts are ordered by Kullback–Leibler distance from the world.

Switching. If the world changes — a regime shift, a new phenomenon — the KL distances re-order, the drifts re-order, and the leading theory changes: a model-selection *phase transition* (Figure 4). This is what you wanted to be able to *see*: you watch the three tallies $\log \mathcal{Z}^A, \log \mathcal{Z}^B, \log \mathcal{Z}^C$, and the structure you should be running is the upper envelope — whichever curve is on top right now. The crossings are where paradigm change happens.

The static moves of §§2–6 are how the agent *responds* to this flow. When the data start demanding a degree of freedom no current theory has, the bordered (expanded) model’s Bayes factor turns positive and the agent *adds* the node — your “new” node arriving. When a coupling goes silent, its pruning Bayes factor turns positive and the agent *removes* it. Expansion and reduction are the two controls by which a believer keeps its information matrix aligned with a moving world; the MGF (§5) prices every move; Woodbury (§6) makes the pricing cheap enough to do continually.

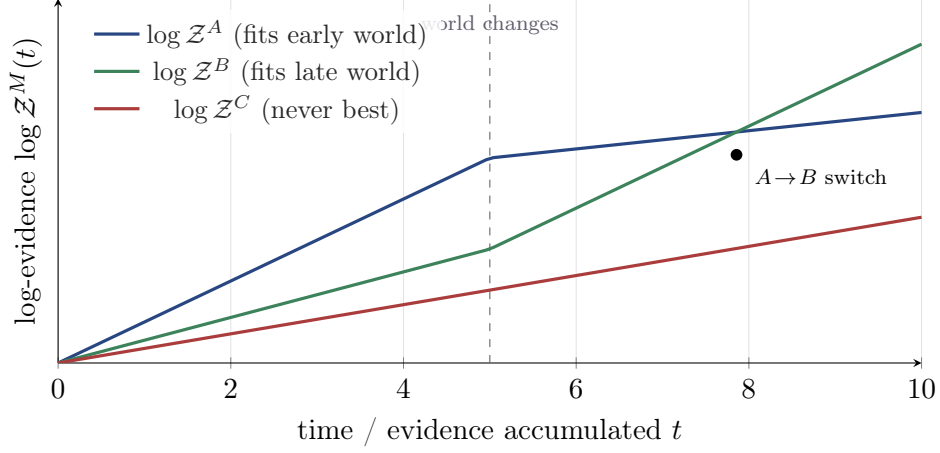
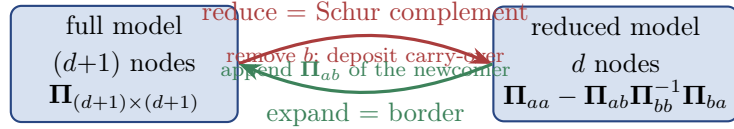


Figure 4: **The dynamics you wanted to see.** Each theory’s log-evidence drifts at a rate set by its KL distance from the world (Eq. 11). Before the regime change A is closest and leads; after it, B ’s drift is steeper and it overtakes A at the crossing (dot). The structure to run is the upper envelope; the crossing is a paradigm shift. C is dominated throughout and would be pruned.

8 Rotating the object: expansion and reduction are adjoint

Stand back and look at the one object from the two sides (Figure 5). Bordering takes a $d \times d$ information matrix to a $(d+1) \times (d+1)$ one by appending the couplings of a newcomer. Schur-complementing takes it back, depositing the newcomer’s influence as carry-over among the survivors. They are inverse operations on the same object: *expansion is bordering*; *reduction is the Schur complement*; and they undo each other up to the ghost the removed node leaves behind. Bordering then immediately Schur-complementing the same block returns Π_{aa} exactly — you get your money back if you remove what you just added. Remove a node the data had *used*, and the Schur term is large: that is precisely an important node, and “importance” is now a number — the size of $\Pi_{ab}\Pi_{bb}^{-1}\Pi_{ba}$, the mass of the off-diagonal block.



price of either direction = the moment generating function $M_{p(\cdot|y)}(\mathbf{t})$
 $\mathbf{t}$ = the prior perturbation of the edit

Figure 5: **One object, two rotations.** Expansion borders the information matrix; reduction Schur-complements it. They are adjoint: removing what you just added returns you exactly, and removing a data-used (“important”) node leaves a large carry-over ghost. Either direction is priced by the same moment generating function evaluated at the edit’s prior perturbation.

9 Reading your sketch in this language

Now the “Bayesian Model Expansion & Reduction” drawing maps cleanly, stroke by stroke.

- **The core net (nodes 1, 2, 3 and their solid edges)** is the incumbent information matrix Π_{aa} — the current paradigm, the block we keep.

- **The dashed “new” node with its $p(1 \mid \text{new}), p(2 \mid \text{new}), \dots$ edges** is a *bordering*: a candidate column Π_{ab} of couplings appended to Π_{aa} (Figure 2). The conditionals you wrote on those dashed edges *are* the entries of that column. Whether the node stays is decided by Eq. (6): compute the bordered model’s Bayes factor and read its sign.
- **“new2”** on the right panel is a *second, competing* expansion — a different candidate column. Expansion is not unique; you propose several borderings and let the evidence (Figure 4) choose the upper envelope.
- **The “IMPORTANT NODE!”** that wires into many others is a hub with a heavy off-diagonal block. Its importance is literal: marginalize it and the carry-over fill-in $\Pi_{ab}\Pi_{bb}^{-1}\Pi_{ba}$ is large (Figure 3); you cannot remove it without massively re-coupling its neighbours. High importance = high carry-over = a large Schur correction = a costly ΔF to prune.
- **“Chris fields correct?”** — the question of whether the right primitive is a node that participates in conditionals across the net — is, in this language, the question of where to place the heavy off-diagonal mass: is the proposed important node’s column Π_{ab} the one the evidence actually supports? That is answerable, and the apparatus above is how: border it, score it by Eq. (6), watch its drift by Eq. (11).

So your picture and BMR are the same picture. Your drawing is the *expansion* half — a new node reaching into the network through conditionals. BMR is the *reduction* half — the Schur complement that takes it back and the moment generating function that prices the trip. Expansion borders; reduction Schur-complements; learning adds Fisher information; the MGF keeps the books; Woodbury makes it cheap; and the upper envelope of the competing log-evidences, drifting and crossing as the world moves, is the paradigm you should be holding.

A The carry-over matrix (Schur complement), slowly

The carry-over term $\Pi_{ab}\Pi_{bb}^{-1}\Pi_{ba}$ did a lot of work above, so here is the same object built up from nothing, four ways, ending in a worked example with actual numbers. The point to carry away: the Schur complement is just *what is left of one block after you eliminate another*, and the elimination always leaves a residue — that residue is the carry-over.

A.1 What it is: block elimination

Take any block matrix and call the blocks A, B, C, D :

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (12)$$

You want to “eliminate” the D corner the way you eliminate a variable in ordinary Gaussian elimination — use D to clear the block above and beside it. Doing that by block row/column operations factors M exactly:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & BD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} M/D & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} I & 0 \\ D^{-1}C & I \end{pmatrix}, \quad \boxed{M/D = A - BD^{-1}C} \quad (13)$$

The middle factor is block-diagonal: D sits in one corner, and in the other corner sits $M/D = A - BD^{-1}C$, *the Schur complement of D in M* . It is literally “the A block, corrected for everything that ran through D .” Two consequences fall out of Eq. (13) by taking determinants and inverting the triangular factors:

$$\det M = \det D \cdot \det(M/D), \quad (14)$$

$$(M^{-1})_{aa} = (M/D)^{-1}. \quad (15)$$

Equation (14) is why *evidence factorises*: the log-determinant of the information matrix splits into the part the removed block carries and the part the survivors carry. Equation (15) says the top-left block of the *inverse* is the inverse of the Schur complement — the fact we use next.

A.2 What it means: marginalize versus condition

Put $M = \mathbf{\Pi}$, the joint precision of (φ_a, φ_b) as in Eq. (3). Now two different “removals” of φ_b pick out two different blocks, and this is the crux you said you wanted nailed down:

	marginalize φ_b (integrate out)	condition on φ_b (observe)
precision $\mathbf{\Pi}$ form	$\mathbf{\Pi}_{aa} - \mathbf{\Pi}_{ab}\mathbf{\Pi}_{bb}^{-1}\mathbf{\Pi}_{ba}$ (Schur)	$\mathbf{\Pi}_{aa}$ (just the block)
covariance $\mathbf{\Sigma}$ form	$\mathbf{\Sigma}_{aa}$ (just the block)	$\mathbf{\Sigma}_{aa} - \mathbf{\Sigma}_{ab}\mathbf{\Sigma}_{bb}^{-1}\mathbf{\Sigma}_{ba}$ (Schur)

Read the table across and a clean duality appears: *the Schur complement is the cost of the “hard” operation in each coordinate system*. In precision coordinates, conditioning is free (you just keep the $\mathbf{\Pi}_{aa}$ block — because the joint exponent already separates that way) and *marginalizing* costs the Schur complement. In covariance coordinates it is exactly reversed. The reason marginalizing is the expensive one in precision form is that integrating out a variable forces you to account for everything that variable was holding together — and that accounting is $\mathbf{\Pi}_{ab}\mathbf{\Pi}_{bb}^{-1}\mathbf{\Pi}_{ba}$.

A.3 Where it comes from: completing the square

This is the derivation worth remembering, because it shows the carry-over being *created*. The joint log-density has the quadratic

$$-\frac{1}{2} \begin{pmatrix} \varphi_a \\ \varphi_b \end{pmatrix}^\top \begin{pmatrix} \mathbf{\Pi}_{aa} & \mathbf{\Pi}_{ab} \\ \mathbf{\Pi}_{ba} & \mathbf{\Pi}_{bb} \end{pmatrix} \begin{pmatrix} \varphi_a \\ \varphi_b \end{pmatrix} = -\frac{1}{2} \left[\varphi_a^\top \mathbf{\Pi}_{aa} \varphi_a + 2\varphi_a^\top \mathbf{\Pi}_{ab} \varphi_b + \varphi_b^\top \mathbf{\Pi}_{bb} \varphi_b \right]. \quad (16)$$

To marginalize φ_b we integrate it out, and to integrate a Gaussian we complete the square in φ_b . The two φ_b terms combine as

$$\varphi_b^\top \mathbf{\Pi}_{bb} \varphi_b + 2\varphi_a^\top \mathbf{\Pi}_{ab} \varphi_b = \underbrace{(\varphi_b + \mathbf{\Pi}_{bb}^{-1} \mathbf{\Pi}_{ba} \varphi_a)^\top \mathbf{\Pi}_{bb} (\varphi_b + \mathbf{\Pi}_{bb}^{-1} \mathbf{\Pi}_{ba} \varphi_a)}_{\text{integrates away to a constant}} - \varphi_a^\top \mathbf{\Pi}_{ab} \mathbf{\Pi}_{bb}^{-1} \mathbf{\Pi}_{ba} \varphi_a. \quad (17)$$

The shifted-square term, integrated over φ_b , gives just a normalising constant. What *stays behind*, glued to φ_a , is the second piece. So the marginal of φ_a has quadratic

$$-\frac{1}{2} \varphi_a^\top (\mathbf{\Pi}_{aa} - \mathbf{\Pi}_{ab} \mathbf{\Pi}_{bb}^{-1} \mathbf{\Pi}_{ba}) \varphi_a, \quad (18)$$

and there it is: *the Schur complement is the residue that completing the square cannot remove*. The carry-over $\mathbf{\Pi}_{ab} \mathbf{\Pi}_{bb}^{-1} \mathbf{\Pi}_{ba}$ is precisely the imprint φ_b leaves on φ_a when you push it out of the model — not a modelling choice, a remainder term.

A.4 A worked example you can see

Take the smallest interesting case: a *star*. Two outer nodes 1, 2 each couple to a central hub b , and crucially 1 and 2 have *no direct edge* between them ($\mathbf{\Pi}_{12} = 0$). In the ordering $(1, 2, b)$, with self-precisions 2 and unit couplings to the hub,

$$\mathbf{\Pi} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad \mathbf{\Pi}_{aa} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{\Pi}_{ab} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{\Pi}_{bb} = 2, \quad \mathbf{\Pi}_{ba} = (1 \quad 1). \quad (19)$$

Marginalize the hub b . The carry-over matrix is a rank-one outer product (because b is one node):

$$\mathbf{\Pi}_{ab}\mathbf{\Pi}_{bb}^{-1}\mathbf{\Pi}_{ba} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}, \quad (20)$$

and the surviving precision is

$$\mathbf{\Pi}_{aa} - \mathbf{\Pi}_{ab}\mathbf{\Pi}_{bb}^{-1}\mathbf{\Pi}_{ba} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{pmatrix}. \quad (21)$$

Two things happened (Figure 6). **An edge was born:** the off-diagonal went from 0 to -0.5 — nodes 1 and 2, which had no direct coupling, are now coupled, purely because they both leaned on b . And **each node lost precision:** the diagonal fell $2 \rightarrow 1.5$, because the hub was helping pin them and now it is gone. The induced coupling is $\mathbf{\Pi}_{1b}\mathbf{\Pi}_{bb}^{-1}\mathbf{\Pi}_{b2} = c_1c_2/d$ — the product of the two couplings to the hub, divided by the hub’s own precision. A hub that many nodes lean on hard (large $\mathbf{\Pi}_{ab}$) leaves a large, dense induced block: *that* is what makes a node “important,” and why an important node is expensive to remove.

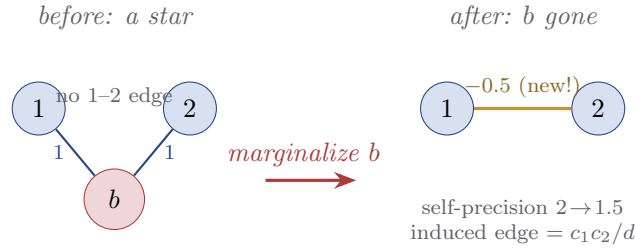


Figure 6: **The carry-over, with numbers.** Marginalizing the hub b of a star induces a direct edge (-0.5) between its two formerly unconnected neighbours and drops their self-precision ($2 \rightarrow 1.5$). The induced coupling is the product of the couplings to b over b ’s own precision — big for a heavily-leaned-on hub. This is the carry-over matrix $\mathbf{\Pi}_{ab}\mathbf{\Pi}_{bb}^{-1}\mathbf{\Pi}_{ba}$ doing its one job.

A.5 The name the numerical analysts use: fill-in

That born edge has a name in sparse linear algebra: *fill-in*. When Gaussian elimination (or Cholesky) eliminates a variable, it connects all of that variable’s neighbours to each other — new nonzeros appear where the original matrix had zeros. The Schur complement *is* one elimination step, and the carry-over *is* its fill-in. This buys two things for the paradigm story:

- **Importance = fill-in.** A node whose elimination produces a lot of fill-in is one whose removal re-wires the whole network — exactly the heavy-Schur-correction “important node” of §9. Belt nodes eliminate cheaply (few neighbours, little fill-in); core hubs eliminate expensively (many neighbours, dense fill-in).
- **Elimination order is belt-first.** Numerical analysts choose the elimination ordering that minimises fill-in (minimum-degree and nested dissection orderings). For a core/belt network that ordering is precisely *belt-first*, *core-last* — which is the same belt-first, core-last sequence the structural reorganisation follows in the population dynamics. The cheapest way to take a paradigm apart and the order in which a community actually takes it apart are the same order, and for the same reason: you do the low-fill-in moves first.

Finally, tie it back to the paper’s carry-over cost. The propagating revision cost $C(v)$ written there as “local KL plus precision-weighted downstream KLs, summed over the cascade” is the *series-expansion* reading of the same object: iterating the Schur elimination one neighbour

at a time generates exactly that sum, and the closed form of the whole cascade is the single matrix $\mathbf{\Pi}_{ab}\mathbf{\Pi}_{bb}^{-1}\mathbf{\Pi}_{ba}$. The cascade picture and the Schur picture are the same carry-over, told step-by-step versus all-at-once.

B The tilt, and why the moment generating function is its price

You said the tilt is the part you cannot see. Let us draw it. There is really one move and one picture: *multiply a density by an exponential, and renormalize*. Everything else — adding evidence, removing a coupling, scoring a model — is that move, and the MGF is the bill it leaves.

B.1 The move: multiply by an exponential, then renormalize

Start with a belief $p(\boldsymbol{\varphi})$ and a “tilt” $e^{\mathbf{t}^\top \boldsymbol{\varphi}}$. This factor is a wager: it pays off exponentially more for large $\mathbf{t}^\top \boldsymbol{\varphi}$, so multiplying by it up-weights one side of your belief and down-weights the other (Figure 7, left). The product $p(\boldsymbol{\varphi}) e^{\mathbf{t}^\top \boldsymbol{\varphi}}$ is no longer a probability density — it does not integrate to one — so you divide by whatever it *does* integrate to. That number is the MGF, and the renormalized object is the *tilted* density:

$$p_t(\boldsymbol{\varphi}) = \frac{p(\boldsymbol{\varphi}) e^{\mathbf{t}^\top \boldsymbol{\varphi}}}{M_p(\mathbf{t})}, \quad M_p(\mathbf{t}) = \int p(\boldsymbol{\varphi}) e^{\mathbf{t}^\top \boldsymbol{\varphi}} d\boldsymbol{\varphi} = \mathbb{E}_p[e^{\mathbf{t}^\top \boldsymbol{\varphi}}]. \quad (22)$$

So the MGF is, quite literally, *the area you sweep up when you tilt* — the total mass of the wagered density. To *add* the wager to your model is to pay that area; to *remove* it is to tilt back by $-\mathbf{t}$ and divide it out.

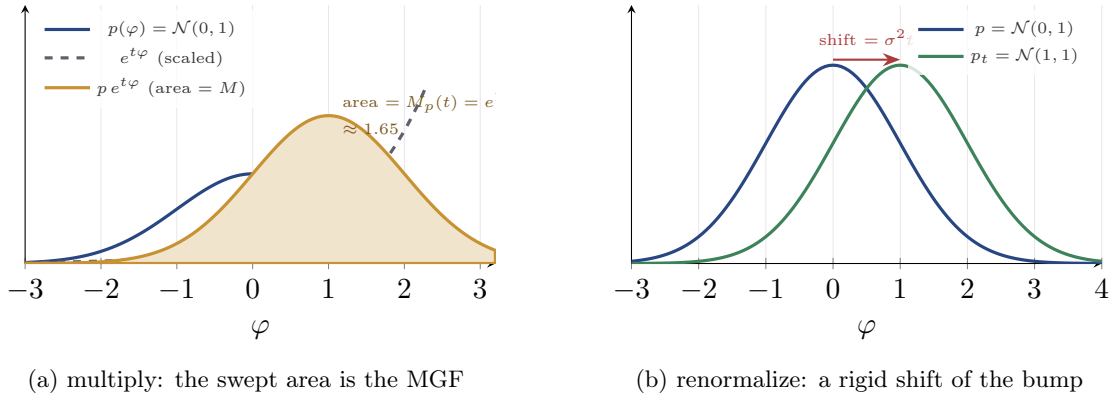


Figure 7: **The tilt, drawn.** (a) Multiplying the belief $p = \mathcal{N}(0, 1)$ by $e^{t\varphi}$ (here $t = 1$) up-weights the right and makes the un-normalized product (amber); the *area* under that product is the moment generating function $M_p(t) = e^{1/2} \approx 1.65$ — the evidence you pay for the wager. (b) Renormalizing gives back a genuine density p_t , which for a Gaussian is just the original *rigidly shifted* by $\sigma^2 t$, same width. To remove the wager you tilt by $-t$ and shift back.

B.2 For a Gaussian, the tilt is a rigid shift

Complete the square in Eq. (22) with $p = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and you find the tilt does something very simple — it slides the mean and leaves the shape alone:

$$p_t = \mathcal{N}(\boldsymbol{\mu} + \boldsymbol{\Sigma} \mathbf{t}, \boldsymbol{\Sigma}), \quad M_p(\mathbf{t}) = \exp\left(\boldsymbol{\mu}^\top \mathbf{t} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}\right), \quad K_p(\mathbf{t}) = \boldsymbol{\mu}^\top \mathbf{t} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}. \quad (23)$$

The mean moves by $\boldsymbol{\Sigma} \mathbf{t}$: *the more uncertain you are, the more a given wager moves you* (a broad belief is easily tilted; a sharp one barely budes). That is the whole content of Figure 7(b), and it is why high-variance — low-precision — beliefs are the ones the data, or a neighbour, can shift.

B.3 The CGF is the ledger; its derivatives are the moments

Take logs: $K_p(\mathbf{t}) = \log M_p(\mathbf{t})$, the cumulant generating function, is the actual ledger. For one parameter it is the parabola $K(t) = \mu t + \frac{1}{2}\sigma^2 t^2$ (Figure 8), and three things are worth reading off it:

$$K(0) = 0, \quad K'(0) = \mu, \quad K''(0) = \sigma^2. \quad (24)$$

At no tilt there is no cost. The *slope at the origin is the mean*; the *curvature is the variance*. This is what “moment generating” means made visible: differentiate the ledger and the moments fall out. More: $K'(\mathbf{t}) = \mathbb{E}_{p_t}[\varphi]$ is the mean of the *tilted* distribution, so as you crank the field $\mathbf{t}$ the slope of the ledger traces where the tilted bump has moved to. And the height $K(\mathbf{t})$ at a chosen tilt is simply the log-evidence you pay for that edit. One curve holds the price, the mean, and the variance at once.

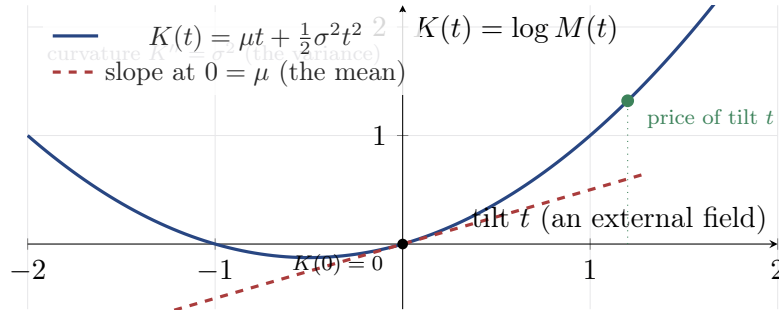


Figure 8: **The ledger.** The cumulant generating function $K(\mathbf{t})$ — for a Gaussian a parabola — prices every tilt. It passes through the origin (no tilt, no cost); its slope there is the mean; its curvature is the variance; and its height at a chosen tilt $\mathbf{t}$ is the log-evidence that edit costs. Differentiating the ledger generates the moments — the literal meaning of “moment generating.”

B.4 Three ways to read the same tilt

The picture is robust because it is the same object physicists, statisticians, and probabilists each met under a different name.

- **Change of measure (statistics).** The tilted density relative to the original is $\frac{dp_t}{dp}(\varphi) = \frac{e^{\mathbf{t}^\top \varphi}}{M_p(\mathbf{t})}$ — a Radon–Nikodym derivative. Tilting is re-weighting one measure into another; importance sampling is exactly this re-weight. And this is why BMR is an MGF: the prior ratio $\tilde{p}/p$ in Eq. (6) is a tilt factor, so $\tilde{\mathcal{Z}}/\mathcal{Z} = \mathbb{E}_{p(\cdot|\mathbf{y})}[\tilde{p}/p]$ is the posterior’s MGF at that tilt — model scoring *is* measuring the area you sweep when you re-weight your posterior by the proposed prior change.
- **Laplace transform (analysis).** $M_p(\mathbf{t}) = \int e^{\mathbf{t}^\top \varphi} p(\varphi) d\varphi$ is the (two-sided) Laplace transform of the density. Tilting-and-reading-the-MGF is taking a transform; the inversion that recovers p from M is the hard direction — the same “inverse is hard” theme of §6.
- **External field / free energy (physics).** Read $\mathbf{t}$ as an external field that couples linearly to φ . Then $M_p(\mathbf{t}) = Z(\mathbf{t})$ is a partition function, $K_p(\mathbf{t}) = \log Z$ is (minus) a free energy, and its derivatives are the response functions: K' is the magnetization (the mean), K'' is the susceptibility (the variance). *Adding a belief is turning on a field; the evidence you pay is the free-energy change; the moments are how the system responds.* For a project built on free energy, this is the tilt’s home: the MGF ledger and the variational free energy are the same bookkeeping, one in the space of beliefs, the other in the space of fields.

So the slogan of §5 is now a picture you can hold. To add a belief is to tilt the bump (Figure 7); the area you sweep is the MGF; the ledger K (Figure 8) charges you its height and hands you the mean and variance as its slope and curvature; and to remove the belief you tilt back and the area divides out. Expansion, reduction, and learning are all the same hand on the same dial — and the dial is a field, the meter is a free energy.